

Beliebige Basen

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1) $52341_6 \rightarrow \text{Basis } 7$

$$\begin{array}{rcl}
 1 \cdot 6^0 & = & 1 \\
 4 \cdot 6^1 & = & 24 \\
 3 \cdot 6^2 & = & 108 \\
 2 \cdot 6^3 & = & 432 \\
 5 \cdot 6^4 & = & 6480 \\
 \hline
 & & 7045_{10}
 \end{array}$$

$$\begin{array}{rcl}
 7045 : 7 & = & 1006 \quad 3R \\
 1006 : 7 & = & 143 \quad 5R \\
 143 : 7 & = & 20 \quad 3R \\
 20 : 7 & = & 2 \quad 6R \\
 2 : 7 & = & 0 \quad 2R
 \end{array}$$

$$\Rightarrow \underline{26353_7}$$

2) $16BB7_{12} \rightarrow \text{Basis } 4$

$$\begin{array}{rcl}
 7 \cdot 12^0 & = & 7 \\
 11 \cdot 12^1 & = & 132 \\
 11 \cdot 12^2 & = & 1584 \\
 6 \cdot 12^3 & = & 10368 \\
 1 \cdot 12^4 & = & 20736 \\
 \hline
 & & 32827_{10}
 \end{array}$$

$$\begin{array}{rcl}
 32827 : 4 & = & 8206 \quad 3R \\
 8206 : 4 & = & 2051 \quad 2R \\
 2051 : 4 & = & 512 \quad 3R \\
 512 : 4 & = & 128 \quad 0R \\
 128 : 4 & = & 32 \quad 0R \\
 32 : 4 & = & 8 \quad 0R \\
 8 : 4 & = & 2 \quad 0R \\
 2 : 4 & = & 0 \quad 2R
 \end{array}$$

$$\Rightarrow \underline{20000323_4}$$

3) $212370_8 \rightarrow \text{Basis } 11$

$$\begin{array}{rcl}
 0 \cdot 8^0 & = & 0 \\
 7 \cdot 8^1 & = & 56 \\
 3 \cdot 8^2 & = & 192 \\
 2 \cdot 8^3 & = & 1024 \\
 1 \cdot 8^4 & = & 4096 \\
 2 \cdot 8^5 & = & 65536 \\
 \hline
 & & 70904_{10}
 \end{array}$$

$$\begin{array}{rcl}
 70904 : 11 & = & 6445 \quad 9R \\
 6445 : 11 & = & 585 \quad 10R \rightarrow A \\
 585 : 11 & = & 53 \quad 2R \\
 53 : 11 & = & 4 \quad 9R \\
 4 : 11 & = & 0 \quad 4R
 \end{array}$$

$$\Rightarrow \underline{492A9_{11}}$$

4) $3,242112_5 \rightarrow \text{Basis 15}$

$$3_5 = 3_{10} = 3_5$$

$$2 = \frac{45157}{5^7} = \underline{0,5780096}$$

$$2 \cdot 5^0 = 2$$

$$1 \cdot 5^1 = 5$$

$$1 \cdot 5^2 = 25$$

$$1 \cdot 5^3 = 125$$

$$2 \cdot 5^4 = 1250$$

$$4 \cdot 5^5 = 12500$$

$$2 \cdot 5^6 = \underline{31250}$$

$$45157$$

$$0,5780096 \cdot 15 = 8,670144 \rightarrow 8$$

$$0,670144 \cdot 15 = 10,05216 \rightarrow A$$

$$0,05216 \cdot 15 = 0,7824 \rightarrow 0$$

$$0,7824 \cdot 15 = 11,736 \rightarrow B$$

$$0,736 \cdot 15 = 11,04 \rightarrow B$$

$$0,04 \cdot 15 = 0,6 \rightarrow 0$$

$$0,6 \cdot 15 = 9 \rightarrow 9$$

$$3,242112_5 = 3,8A0DB09$$

5) $2,3_4 \rightarrow \text{Basis 3}$

$$2 \cdot 4^0 = 2_{10}$$

$$0,3 \rightarrow 3 \cdot 4^0 = 3$$

$$2 = \frac{3}{4} = 0,75$$

$$\Rightarrow 2,3_4 = 2,75_{10}$$

$$2:3 = 0 \quad 2R$$

$$0,75 \cdot 3 = 2,25 \quad 2$$

$$0,25 \cdot 3 = 0,75 \quad 0$$

$$0,75 \cdot 3 = 2,25 \quad 2$$

⋮

$$2,3_4 = 2,\overline{202}_3$$

6) $50D, 68_{16} \rightarrow \text{Basis } 9$

$$\begin{array}{l} D: 13 \cdot 16^0 = 13 \\ 0: 0 \cdot 16^1 = 0 \\ 5: 5 \cdot 16^2 = 1280 \\ \hline 1293 \end{array}$$

$$\begin{array}{l} 8 \cdot 16^0 = 8 \\ 6 \cdot 16^1 = 96 \\ \hline 104 \end{array}$$

$$z = \frac{104}{16^2} = 0,40625$$

$$\Rightarrow 50D, 68_{16} = 1239,40625_{10}$$

$$\begin{array}{ll} 1239 : 9 = 143 & 6R \\ 143 : 9 = 15 & 8R \\ 15 : 9 = 1 & 6R \\ 1 : 9 = 0 & 1R \end{array}$$

$$\begin{array}{ll} 0,40625 \cdot 9 = 3,65625 & \rightarrow 3 \\ 0,65625 \cdot 9 = 5,90625 & \rightarrow 5 \\ 0,90625 \cdot 9 = 8,15625 & \rightarrow 8 \\ 0,15625 \cdot 9 = 1,40625 & \rightarrow 1 \\ 0,40625 \cdot 9 = 3,65625 & \rightarrow 3 \\ & \vdots \end{array}$$

$$50D, 68_{16} \rightarrow 1686, \overline{35813}_9$$